Chumeng Liang

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Research Interest

I study **language models with new architecture**, with my current focus on **continuous diffusion language models** and **speculative decoding**, both of which have the potential to enable fast language generation.

Highlights

LangFlow: Continuous Diffusion Rivals Discrete in Language Modeling

First continuous diffusion language model with better sample quality to masked diffusion on the GPT-2 scale.

UBTree: Parallel Tree Drafting via Unigram and Bigram Models for Speculative Decoding

State-of-the-art single-request speculative decoding method.

CBD: Scaling Continuous Block Diffusion (To come)

Scaling likelihood-based continuous diffusion language modeling with elucidated designs to 8B from scratch.

Education

University of Illinois, Urbana-Champaign, Computer Science, *PhD Candidate* Aug. 2025 – Present

Advisor: Jiaxuan You, Ge Liu

University of Southern California, Computer Science, *MS* Aug. 2023 – May 2025

Advisor: Jiaxuan You, Ge Liu

Shanghai Jiao Tong University, Computer Science and Technology, *BEng* Sep. 2018 – Jun. 2022

Advisor: Yang Hua, Guanjie Zheng

Work Experience

Pretraining Continuous Diffusion Language Models at Scales May. 2026 – Aug. 2026

Mentor: Jianguo Li LLADA Pretraining Team, Inclusion AI

- Re-designed likelihood-based continuous language diffusion for better maximum likelihood training.
- Scaled continuous language diffusion to 8B parameters under different network architecture.

Parallel Tree-based Speculative Decoding for LLMs May. 2026 – Aug. 2026

Mentor: Jianguo Li LLADA Pretraining Team, Inclusion AI

- Developed speculative draft models with both parallel and tree-based decoding for Ling3-Flash-124B.

Publications

- *UBTree: Parallel Tree Drafting via Unigram and Bigram Models for Speculative Decoding*
  Chumeng Liang*, Linxuan Wang*, Xinyu Peng*, Huabin Liu, Xiaoxu Guo, Yuxin Chen, Ge Liu, Guang Lin, Qifan Song, Jianguo Li
  In submission to **ICLR 2027**
- *LangFlow: Continuous Diffusion Rivals Discrete in Language Modeling*
  Yuxin Chen*, **Chumeng Liang***, Hangke Sui*, Ruihan Guo, Chaoran Cheng, Jiaxuan You, Ge Liu
  Accepted by **NeurIPS 2026**
- *Adversarial Example Does Good: Preventing Painting Imitation from Diffusion Models via Adversarial Examples*
  Chumeng Liang*, Xiaoyu Wu*, Yang Hua, Jiaru Zhang, Yiming Xue, Tao Song, Zhengui Xue, Ruhui Ma, Haibing Guan
  Accepted by **ICML 2023** (**Oral**)
- *Targeted Attack Improves Protection against Unauthorized Diffusion Customization*
  Boyang Zheng*, **Chumeng Liang***, Xiaoyu Wu
  Accepted by **ICLR 2025** (**Spotlight**)

- *Evaluating LLM-Generated Diagrams as Graphs*
  Chumeng Liang, Jiaxuan You
  Accepted by **EMNLP 2025 Main**
- *CBLab: Supporting the Training of Large-scale Traffic Control Policies with Scalable Traffic Simulation*
  Chumeng Liang, Zherui Huang, Yicheng Liu, Zhanyu Liu, Guanjie Zheng, Hanyuan Shi, Kan Wu, Yuhao Du, Fuliang Li, Zhenhui Jessie Li
  Accepted by **SIGKDD 2023**
- *Toward Effective Protection against Diffusion-based Mimicry through Score Distillation*
  Haotian Xue, **Chumeng Liang***, Xiaoyu Wu*, Yongxin Chen
  Accepted by **ICLR 2024**
- *Cgi-dm: Digital Copyright Authentication for Diffusion Models via Contrasting Gradient Inversion*
  Xiaoyu Wu, Yang Hua, **Chumeng Liang**, Jiaru Zhang, Hao Wang, Tao Song, Haibing Guan
  Accepted by **CVPR 2024**
- *Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization*
  Jiajun Fan, Shuaike Shen, Chaoran Cheng, Yuxin Chen, **Chumeng Liang**, Ge Liu
  Accepted by **ICLR 2025**
- *FDTI: Fine-grained Deep Traffic Inference with Roadnet-enriched Graph*
  Zhanyu Liu, **Chumeng Liang**, Guanjie Zheng
  Accepted by **ECML-PKDD 2023**
- *Mist: Towards Improved Adversarial Examples for Diffusion Models*
  Chumeng Liang*, Xiaoyu Wu*
  Technical Report

* denotes equal contribution

Service & Teaching

- **Reviewer**: ICML (2025, 2026), NeurIPS (2024, 2025, 2026), ICLR (2025, 2026, 2027), AISTATS (2025)
- **Teaching Assistant**: CS 105: Introduction to Computing, University of Illinois Urbana Champaign, 2026
- **Teaching Assistant**: DSCI352: Applied Machine Learning and Data Mining, University of Southern California, 2024